

Bonus: Course Enrollment Directory with `std::map`

Instructions

You must write a C++ program that simulates a simple course enrollment system.

Learning goals

- Create and use a `map<int, string>`
- Insert elements using `operator[]`
- Traverse a map using an iterator
- Access key-value pairs using `it->first` and `it->second`
- Observe that maps store keys in sorted order

Scenario

You are given a list of students enrolling in a course. Each student has a unique ID and a name.

Your task is to build and display the enrollment list.

Task

1. You must create an empty map called `enrollment` that maps student IDs to names.
2. You must add the following students using `operator[]` (in this order):
 - `3003 → "Charlie"`
 - `3001 → "Alice"`
 - `3004 → "Diana"`
 - `3002 → "Bob"`
3. You must print the enrollment list using an iterator.
4. You must number the students starting from 1 when printing.
5. You must include a comment in your code that answers this question:

- Why is "Alice" printed before "Charlie" even though she was added later?
6. You must create a second map `lateEnrollment` using an initializer list:
- 4002 $\rightarrow$ "Evan"
 - 4001 $\rightarrow$ "Fiona"
7. You must print the second map in the same format.

Expected output

Enrollment list:

1. ID 3001: Alice
2. ID 3002: Bob
3. ID 3003: Charlie
4. ID 3004: Diana

Late enrollment:

1. ID 4001: Fiona
2. ID 4002: Evan

Starter code

```
#include <iostream>
#include <map>
#include <string>
using namespace std;

int main()
{
    // Create enrollment map

    // Add students

    // Print enrollment list with numbering

    // Create lateEnrollment map

    // Print late enrollment
```

```
    return 0;
}
```

Submission

You must submit:

1. A working program hosted on OneCompiler.com.
2. The URL to your solution.

Your submission must:

- Compile and run without errors.
- Produce output that matches the format shown above.
- Include your answer (as a comment) explaining the ordering of the map.

Example submission format:

URL: <https://onecompiler.com/cpp/xxxxxxx>

Sample output:

Enrollment list:

1. ID 3001: Alice
2. ID 3002: Bob
3. ID 3003: Charlie
4. ID 3004: Diana

Late enrollment:

1. ID 4001: Fiona
2. ID 4002: Evan